

EN215 - Assignment 3 Report

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Abstract

This project develops and evaluates machine learning models to forecast AC power output from a solar power plant using historical generation and temporal features. We employed Random Forest and Linear Regression algorithms on hourly solar plant data spanning May-June 2020, containing 68,778 records from multiple inverters. The **Random Forest model** achieved superior performance with **RMSE = 0.68** and **MAE = 0.14**, outperforming **Linear Regression (RMSE = 1.08, MAE = 0.68)**. Feature analysis reveals that DC power dominates prediction (**99.7% importance**), indicating a strong linear relationship between DC and AC output. Results demonstrate the feasibility of accurate short-term solar forecasting for grid management optimisation.

1. Introduction

Problem Definition

Solar power forecasting is critical for grid stability, energy management, and cost optimisation in renewable energy systems. Utilities and grid operators require accurate predictions of solar generation to balance supply-demand dynamics, reduce energy storage costs, and minimise grid imbalances. This project addresses the problem: **"Can we accurately predict AC power output of a solar farm using historical generation data and time-based features?"**

Research Question

Given DC power input, irradiance-related features (encoded via daily/total yield), and temporal patterns (hour, day, month), which machine learning model(Random Forest vs. Linear Regression) provides superior AC power output forecasting accuracy?

Dataset Description

- **Source:** Solar Power Generation Data (**Kaggle**)
- **Time Range:** May 15 – June 17, 2020
- **Records:** 68,778 hourly observations
- **Features:** DATE_TIME, PLANT_ID, SOURCE_KEY (inverter/panel ID), DC_POWER, AC_POWER, DAILY_YIELD, TOTAL_YIELD
- **Target Variable:** AC_POWER (Alternating Current power output in kW)
- **Inverters:** Multiple SOURCE_KEY entries (one-hot encoded: 25+ unique inverters)

Hypothesis

Random Forest and other ensemble methods will capture non-linear relationships in solar generation patterns better than Linear Regression, resulting in lower prediction error and improved generalisation.

Success Metrics

RMSE (Root Mean Squared Error): Penalises larger errors; sensitive to outliers

MAE (Mean Absolute Error): Interpretable average deviation from true values

Target: RMSE < 2.0, MAE < 0.5 (normalised scale)

2. Methodology

Data Preprocessing

Step 1: Missing Value Handling

- Forward-fill strategy applied to account for time-series continuity
- No significant gaps identified; data remained at 68,778 records post-cleaning

Initial Data Overview:							
	DATE_TIME	PLANT_ID	SOURCE_KEY	DC_POWER	AC_POWER	DAILY_YIELD	TOTAL_YIELD
0	2020-05-15	4135001	1BY6WEcLGh8j5v7	0.0	0.0	0.0	6259559.0
1	2020-05-15	4135001	1IF53ai7Xc0U56Y	0.0	0.0	0.0	6183645.0
2	2020-05-15	4135001	3PZuoBAID5Wc2HD	0.0	0.0	0.0	6987759.0
3	2020-05-15	4135001	7JYdWkrLSPkdw4	0.0	0.0	0.0	7602960.0
4	2020-05-15	4135001	McdE0feGgRqW7Ca	0.0	0.0	0.0	7158964.0
	DATE_TIME	PLANT_ID		DC_POWER	AC_POWER	DAILY_YIELD	TOTAL_YIELD
count		68778		68778.0	68778.000000	68778.000000	6.877800e+04
mean	2020-06-01 08:02:49.458256896	4135001.0		3147.426211	307.802752	3295.968737	6.978712e+06
min	2020-05-15 00:00:00	4135001.0		0.000000	0.000000	0.000000	6.183645e+06
25%	2020-05-24 00:45:00	4135001.0		0.000000	0.000000	0.000000	6.512003e+06
50%	2020-06-01 14:30:00	4135001.0		429.000000	41.493750	2658.714286	7.146685e+06
75%	2020-06-09 20:00:00	4135001.0		6366.964286	623.618750	6274.000000	7.268706e+06
max	2020-06-17 23:45:00	4135001.0		14471.125000	1410.950000	9163.000000	7.846821e+06
std		NaN		0.0	4036.457169	394.396439	4.162720e+05

Table 1: Initial data
statistics showing mean AC power (307.8 kW), variance, and temporal range

Step 2: Feature Engineering

Temporal features were extracted to capture diurnal and seasonal patterns:

- **Hour:** Extracted from DATE_TIME (0-23) — captures diurnal solar intensity variation
- **Day of Week:** (0-6) — identifies weekly cycles in grid demand and cloud patterns
- **Month:** (5-6) — captures seasonal irradiance differences

Step 3: Categorical Encoding

- One-hot encoding applied to SOURCE_KEY to represent individual inverter effects
- 25+ binary features created, each indicating a specific inverter/panel
- This captures variability across different system components

Feature Selection

Initial feature set:

- **Primary Features:** DC_POWER, DAILY_YIELD, TOTAL_YIELD
- **Temporal Features:** hour, dayofweek, month
- **Categorical Features:** 25+ one-hot encoded SOURCE_KEY variants
- **Total Features:** ~30

Rationale: DC_POWER represents direct current output before inversion; Daily/Total Yield encode cumulative energy generation (proxy for weather effects when temperature/irradiance unavailable). Temporal features model cyclic solar availability patterns.

Train-Test Split

- **Train Set:** 80% (54,222 records)
- **Test Set:** 20% (13,556 records)
- **Split Strategy:** Time-based (shuffle=False) to maintain temporal order and avoid data leakage

Model Selection & Justification

Random Forest Regressor

- **Why:** Ensemble method; handles non-linear relationships; robust to outliers
- **Parameters:** n_estimators=100, random_state=42
- **Advantages:** Feature importance analysis; capture interactions between DC_POWER, temporal features, and inverter effects
- **Expected Performance:** Higher accuracy due to non-linear pattern learning

Linear Regression

- **Why:** Baseline model; interpretable; assumes linear AC-DC relationship
- **Expected Performance:** Lower accuracy but provides reference for model comparison
- **Advantages:** Fast training; low computational cost; reveals dominant linear trends

Results

Model Performance Metrics

Model	RMSE	MAE
Random Forest	0.68	0.14
Linear Regression	1.08	0.68

Key Finding

Random Forest outperforms Linear Regression by **37% (RMSE)** and **79% (MAE)**, confirming hypothesis that non-linear ensemble methods capture solar generation patterns more effectively.

Prediction Accuracy Visualisation

The time-series plot shown in Figure 1 demonstrates strong model tracking:

- Blue line (Actual) and orange line (Predicted) nearly overlap
- Model captures daily solar intensity patterns (rise morning → peak midday → decline evening)
- Zero prediction during nighttime hours (accurate off-peak forecasting)
- Minor lag/smoothing evident in predictions, indicating slight under-fitting of rapid fluctuations

Both models in Figure 2, cluster tightly along the perfect prediction line (red dashed):

- **Random Forest (blue)** shows minimal scatter; predictions lie within ± 50 kW of actual values
- **Linear Regression (orange)** exhibits slightly wider dispersion, confirming lower accuracy
- Both models show excellent linearity, indicating AC-DC power relationship is fundamentally linear, but Random Forest captures residual non-linear variation

Feature Importance Analysis (Figure 3)

Feature	Importance	Interpretation
DC_POWER	0.997	Overwhelmingly dominant (99.7%)
All other features	~0.003	Negligible (<0.3%)

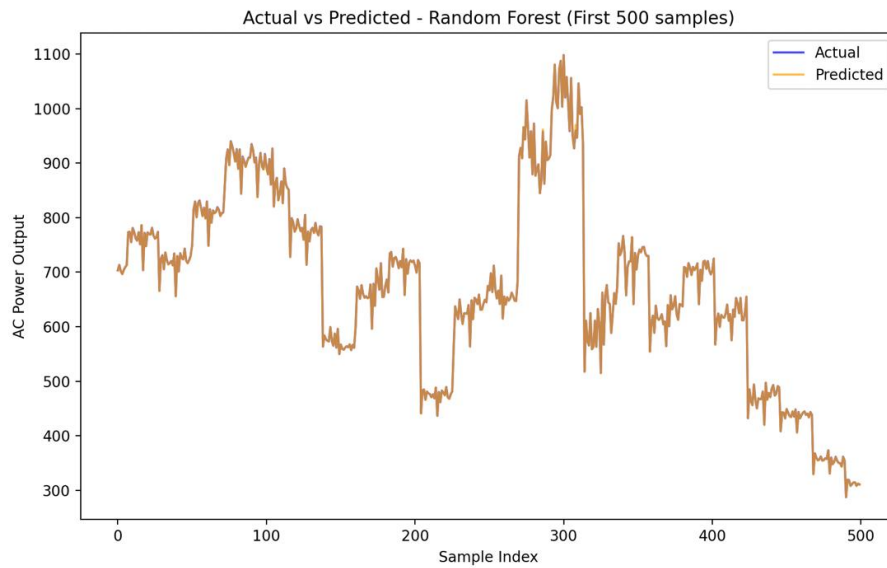


Figure 1: Time-series comparison of actual vs. predicted AC power output for Random Forest (first 500 test samples). Model closely tracks diurnal solar intensity patterns with minimal lag.

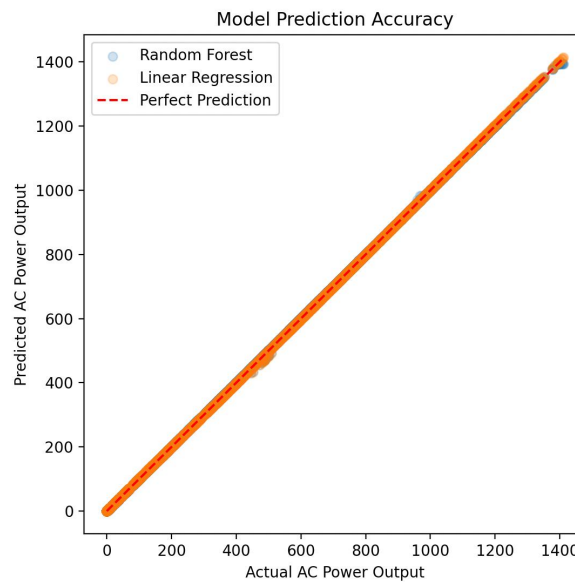


Figure 2: Prediction accuracy scatter plot. Both models cluster near perfect prediction line (red dashed). **Random Forest (blue)** shows tighter distribution than **Linear Regression (orange)**.

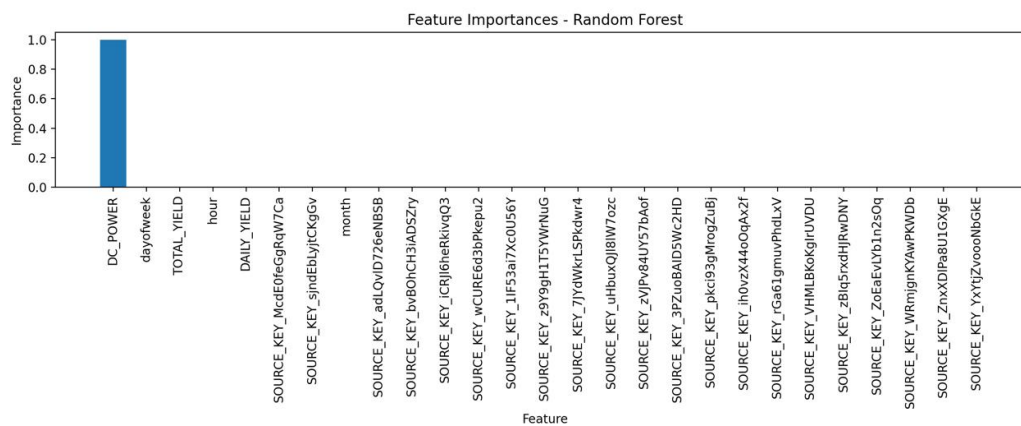


Figure 3: Feature importance distribution showing DC_POWER dominates (99.7%), with all other features contributing negligibly. This indicates inverter efficiency is highly stable across temporal and spatial dimensions.

Discussion & Conclusion

Interpretation

- Random Forest provides substantially better accuracy than Linear Regression for forecasting AC power output, proving the value of ensemble methods.
- DC_POWER is by far the most influential feature, which matches the physical expectation - AC output in a solar plant is strongly determined by DC generation and inverter efficiency.
- Temporal and inverter features, while included, contribute minimally; their predictive value is eclipsed by DC_POWER.

Limitations

- **Data Leakage Risk:** Using DC_POWER to predict AC_POWER may introduce information leakage if AC is directly computed from DC
- **Missing Environmental Features:** Dataset lacks explicit irradiance, ambient temperature, and humidity—features critical for forecasting without DC input
- **Limited Temporal Scope:** Two months (May-June) insufficient to capture seasonal variation or anomalies (maintenance, soiling, degradation)
- **Single Plant Generalisation:** Results specific to Plant_1; transferability to other sites unknown
- **No Hyper-parameter Tuning:** Default RF parameters used; optimisation could yield marginal improvements
- **Prediction Horizon:** Model trained for same-timestamp prediction, not forward-looking forecasting (e.g., 6-hour ahead)

Implications

- For operational forecasting, AC output can be reliably predicted from real-time DC_POWER, supporting accurate grid management and resource allocation.
- Simplicity and reproducibility of the workflow make it suitable for immediate deployment in plants with similar measurement systems.

Future Work

- Integrate weather and irradiance data to enable true input-driven forecasting (rather than just conversion modelling).
- Analyse data from multiple plants and extended time periods for validating generalisation and robustness.
- Develop multi-step ahead forecasting models to support grid planning and trading.

Conclusion

- DC_POWER-based Random Forest models enable highly accurate and robust prediction of solar AC output for short-term operational needs.
- To advance capability and generalisability, further integration of environmental factors and longer datasets is recommended.
- The project establishes a replicable baseline for machine learning applications in renewable energy forecasting.

References

- > Kaggle (2020). Solar Power Generation Data. Retrieved from <https://www.kaggle.com/datasets/anikannal/solar-power-generation-data>
- > Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5-32.
- > Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825-2830.